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ORIGINAL PAPER

Exploring the Antecedents of Unethical Pro-organizational Behavior (UPB): A Meta-Analysis

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Abstract
Scholars have paid so much academic attention to UPB in the past decade. However, there is lacking a quantitative review to uncover the relationship between UPB and its antecedents. To address this, we make a meta-analytic review about UPB. Specifically, we propose a theoretical framework of antecedents of UPB and test it using meta-analysis technology ($k = 67$, $n = 20{,}957$). We found moral disengagement, organizational identification, identification with supervisors, leader UPB, ethical judgments, psychological entitlement, transformational leadership, and job satisfaction are positively related to UPB. Moral identity and taking responsibility are negatively related to UPB. Besides, ethical leadership, abusive supervision, and authoritarian leadership have double-edged sword effects on UPB. For ethical leadership, it increases UPB via LMX, on the other hand, it decreases UPB via moral disengagement. For abusive supervision and authoritarian leadership, they increase UPB via moral disengagement and decrease UPB via LMX meanwhile. We discuss the results of the current meta-analysis and put some future research directions.

Keywords Unethical pro-organizational behavior · Meta-analysis · Antecedents

Introduction

Unethical pro-organizational behavior (UPB), which is defined as "actions that are intended to promote the effective functioning of the organization or its members (e.g., leaders) and violate core societal values, mores, laws, or standards of proper conduct" (Umphress & Bingham, 2011), has drawn plentiful of research attention recently. The most widely used measure of UPB is the scale developed by Umphress & Bingham (2011), which captures an employee's self-reported willingness to engage in UPB. In organizational situations, UPB is a phenomenon that often occurs; for example, salesmen hide the faults of goods and sell them to customers, and accountants help the company to conduct financial fraud. Although UPB may be beneficial for organizational short-term profit, it may harm the interests of stakeholders such as employees, customers, and the public, which is detrimental to the long-term interests of the organization (Umphress & Bingham, 2011). Given such negative impacts, researchers have identified a lot of antecedents of UPB (e.g., ethical leadership, Tang et al., 2020; moral disengagement, Chen et al., 2016; and organizational identification, Umphress et al., 2010) from different perspectives.

However, previous studies showed somewhat inconsistent results between UPB and its antecedents. For instance, ethical leadership was found positively related to UPB in some studies (Kalshoven et al., 2016; Lee et al., 2020; Miao et al., 2013) but negatively related to UPB in some other studies (Lian et al., 2020; Tang & Li, 2021). Some qualitative literature reviews noticed the problem (e.g., Mishra et al., 2021; Veetikazhi et al., 2020), but they cannot address this issue sufficiently because they cannot avoid the distorting effects of statistical artifacts (e.g., measurement error and sampling error), nor can they accumulate empirical results to establish the relationships between UPB and its antecedents (Hunter & Schmidt, 2004). The lack of a quantitative review of UPB

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may limit the development of this field. Thus, the first goal of the current study is to make a quantitative review drawing on meta-analysis technology. Meta-analysis enables us to know not only what factors are related to UPB, but also to what extent they are related. Specially, to start, we develop a series of hypotheses between UPB and its antecedents. Then, we collect data from the articles about UPB. Next, we provide a meta-analysis between UPB and its antecedents drawing on the Hunter–Schmidt random-effect meta-analysis method.

Besides, as mentioned earlier, previous studies found inconsistent relationships between ethical leadership and UPB. Statistical artifacts (e.g., sampling error) may lead to this phenomenon. Another possible reason is that there are opposite mechanisms existing in the relationship between ethical leadership and UPB. An early review study (Mishra et al., 2021) identified two major theoretical perspectives in the UPB literature, namely, social cognitive theory (SCT) and social exchange theory (SET). We apply this SCT/SET framework to develop the hypotheses of UPB. Drawing on the SET, we expect that ethical leadership would positively influence UPB via leader–member exchange (LMX); at the same time, drawing on SCT, we expect that ethical leadership would negatively influence UPB via moral disengagement. As such, due to two opposite mechanisms, the true-score correlation between ethical leadership and UPB would be close to zero. Beyond ethical leadership, existing studies also suggest inconsistent findings regarding the relationships between other two leadership styles (i.e., abusive supervision and authoritarian leadership) and UPB. For example, for abusive supervision, Guo et al. (2020) found a negative association between abusive supervision and UPB, whereas Xiong et al. (2021) found a positive linkage; for authoritarian leadership, a prior study revealed a very small correlation ($r = 0.09$) between authoritarian leadership and UPB (Liu et al., 2021). However, these potential double-edged effects have not been empirically tested in an integrated framework. Therefore, the second goal of the present study is to integrate SCT and SET to test the double-edged sword effects of ethical leadership, abusive supervision, and authoritarian leadership on UPB. In particular, *meta-analytic structural equation modeling (MASEM)* technology is used to conduct path analysis. Compare to traditional SEM, MASEM has higher external validity and more sample size as it combined evidence from multiple organizations (Bergh et al., 2016), providing solid evidence to test the mechanisms between these three leadership styles and UPB.

By doing so, we provide both theoretical and methodological contributions to research on the antecedents of UPB. From a methodological level, we used the Hunter-Schmidt method which is the most widely used meta-analytical approach (Aguinis et al., 2010) to calculate true-score correlations between UPB and its antecedents. This effort can help us to justify the inconsistent findings regarding the antecedents of UPB in the previous empirical studies, providing solid evidence for UPB literature. Theoretically, we drew upon SCT and SET and used MASEM technology to test the mediating hypotheses linking ethical leadership, abusive supervision, and authoritarian leadership to UPB. This attempt helps us to empirically justify the double-edged sword effects of these leadership styles on UPB, providing a more complete understanding of the effects—both positive and negative—of ethical leadership, abusive supervision, and authoritarian leadership on UPB. Besides, our research integrated social cognition and social exchange theories to establish a paradoxical perspective on the associations between leadership and UPB, which is consistent with the inherent nature of UPB (i.e., including both unethical elements and pro-organizational elements).

Hypothesis Development

According to the definition of UPB (Umphress & Bingham, 2011), it includes two key components: (1) unethical behavior and (2) pro-organizational behavior. The theoretical underpinnings used to explain the drivers of these two components are different. Specifically, SCT was often used to explain UPB's unethical part; for its pro-organizational part, SET was more widely adopted (Mishra et al., 2021; Umphress et al., 2010). Based on SCT and SET, we develop the following hypotheses to unpack the antecedents of UPB. In addition, we also identified several demographic variables as antecedents of UPB based on the existing literature. Figure 1 is our theoretical model which integrates all the above hypotheses.

Social Cognition Theory

Based on SCT, individual behavior is shaped by her or his self-regulation (usually influenced by personal attributes) toward a specific social context (Wood & Bandura, 1989). In terms of UPB, people equipped with the attributes which are closely bounded up with unethical behaviors (e.g., Machiavellianism, Mishra et al., 2021; psychological entitlement, Lee et al., 2017) are more likely to exhibit actual UPB in some inducible social conditions (e.g., perceived forgiveness climate, Yao et al., 2021). The underlying mechanism is moral disengagement, which is a cognitive process that involves the deactivation of moral self-sanctions (Bandura et al., 1996). Unethical behaviors may happen when people's self-regulatory moral standards break away from socialized moral standards (Bandura et al., 1996; Chen et al., 2016). If these behaviors are reframed by some social factors (e.g., organizational culture) as righteous acts which are necessary for

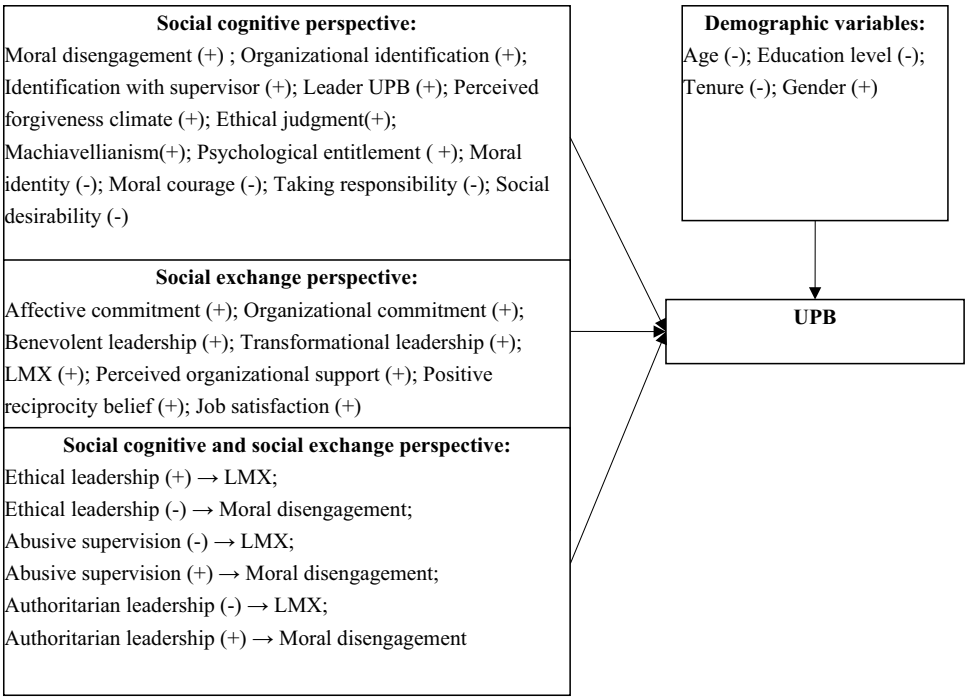


Fig. 1 Theoretical model

the organization's benefits, employees may find proper excuses for their unethical acts, and engage in more UPB in the future (Mishra et al., 2021). Thus, we argue that the social factors which can activate the cognitive process of moral disengagement may induce UPB. Drawing on the theoretical arguments proposed by Bandura et al. (1996), we identify three cognitive mechanisms embedded in the process of moral disengagement.

The first mechanism is *moral justification*. In this process, the unethical act may bring personal and social recognition by portraying it as serving socially worthy or moral purposes (Bandura, 2002). In the workplace, moral justification is reflected by the cognitive process of doing unethical behaviors in the name of organizational goals or benefits. According to this mechanism, we develop the hypotheses between UPB and organizational identification, identification with supervisor, leader UPB, and Machiavellianism. Early studies have found organizational identification (Chen et al., 2016), identification with supervisor (Shaw & Liao, 2020), leader UPB (Lian et al., 2020), and Machiavellianism (Toirova & Baek, 2021) are positively related to UPB.

Organizational Identification

Organizational identification, defined as "a specific form of social identification where the individual defines him or herself in terms of their membership in a particular organization" (Mael & Ashforth, 1992,p 105), may influence the process of moral disengagement, which then leads to UPB. People high in organizational identification tend to see themselves as a part of the organization. Thus, they are willing to do anything that is beneficial to the organization, even it is unethical. They justify their unethical behaviors in the name of pursuing organizational interests.

Identification with Supervisors

Similar to organizational identification, people who identify with their supervisors are also likely to engage in UPB because they can find a plausible excuse for their UPB: the pursuit of goals set up by the supervisors. Usually, finishing the goals issued by the supervisor is consistent with pursuing organizational interests.

Leader's UPB

According to SCT, a leader usually serves as a role model for his or her followers, implying that the followers' cognitions and behaviors should be impacted by their leader (Bandura et al., 1996). When leaders engaged in UPB without being punished, employees could utilize this case to justify their own UPB.

Machiavellianism

Machiavellianism which endorses a morality that prioritizes expediency over ethical principles (Forsyth et al.,

2012) should be a driver of UPB. Individuals who are high in Machiavellianism use aggressive and devious methods to achieve goals without regard for the feelings, rights, and needs of other people (Wilson et al., 1996). They have a utilitarian sense of morality and believe "ends justify the means" (Castille et al., 2016). They justify their behaviors in the name of the organization. As such, they will be willing to use any means necessary to achieve a certain result (even if such a result violates ethical standards).

The second mechanism of moral disengagement is *minimizing or ignoring the consequences of unethical behaviors* (Bandura, 2002; Bandura et al., 1996). Employees are more likely to do unethical behavior because they do not need to shoulder the responsibility of bad consequences caused by unethical behaviors; instead, they engage in less unethical behavior if they need to fully take responsibility. Early studies found ethical judgment (Graham et al., 2019) and perceived forgiveness climate (Graham et al., 2019) are positively related to UPB, while moral identity (Shaw & Liao, 2020), moral courage (Wang & Li, 2019), social desirability (Wang et al., 2018), and taking responsibility (Wang & Li, 2019) are negatively linked to UPB. Based on this mechanism, we will develop the hypotheses between UPB and them.

Ethical Judgment

The ethical judgment of UPB can impact individual's cognitive evaluation of UPB. Self-sanctions are activated when the personal agency for unethical effects is acknowledged (Bandura et al., 1996). However, as people's moral standards could be different, they make different ethical judgments of UPB. Individuals who judge UPB as more ethical may trigger less self-sanction and self-condemnation, and thereby engage in UPB.

Perceived Forgiveness Climate

If an employee works in an organization that is inclusive (i.e., high levels of forgiveness climate), she or he is more likely to do UPB, because their unethical behaviors can be understood and forgiven in such an organizational climate. In other words, employees are very likely to activate the process of moral disengagement, when the negative consequences of unethical behavior can be ignored or minimized by the organization.

Moral Identity

Moral identity is a trait-specific (i.e., morality) definition of the self and reflects people's inner moral standards (Aquino & Reed II, 2002). Minimizing or ignoring the consequences of the unethical act would harm the individuals' moral identity as a moral agency. To keep the consistency of self-concept, people with high levels of moral identity rarely find excuses for unethical behaviors. In other words, they are less likely to activate the process of moral disengagement and engage in UPB.

Moral Courage

Similar to moral identity, moral courage is a personality strength reflecting persistence in personal moral principles and engaging in ethical behaviors (Wang & Li, 2019). Employees with high levels of moral courage would not minimize or ignore the consequences caused by unethical behavior. In other words, the moral disengagement process would not happen. As such, moral courage may restrict UPB.

Taking Responsibility

An important way to disengage morality is to ignore or minimize the negative consequences of unethical behavior (Bandura et al., 1996). In other words, if employees do not shoulder the responsibility for the outcomes of their unethical acts, moral disengagement may be more likely to occur, and they are more easily to do UPB. Conversely, people who take responsibility for unethical behavior are less likely to do UPB.

Social Desirability

Social desirability, which reflects the tendency of an individual to deny socially undesirable actions and to admit to socially desirable ones (Zerbe & Paulhus, 1987), may serve as a hinder to UPB. Employees with a high level of social desirability may not ignore social responsibility as taking responsibility is the expectation of society. A lot of work on UPB regarded social desirability as a control variable because people with a high level of social desirability may report a low level of unethical behavior (e.g., Toirova & Baek, 2021; Umphress et al., 2010), suggesting that social desirability may be negatively related to UPB.

The third mechanism of moral disengagement is *operating on the recipients of unethical acts* (Bandura, 2002; Bandura et al., 1996). Through devaluing and blaming the victims of UPB (Chen et al., 2016), the moral agency would feel lower strength of moral self-sanctions, as the strength of self-sanctions depend partly on the extent to which perpetrators view the people they act unethical behavior (Bandura et al., 1996).

Psychological Entitlement

Drawing on the above mechanism, psychological entitlement may positively relate to UPB. Psychological entitlement is a pervasive sense that one deserves more than others (Campbell et al., 2004). Believing that one deserves more than others, to some extent, reflects the devaluating to other people. Besides, individuals with a high level of psychological entitlement are more likely to blame others (Harvey & Martinko, 2009). As such, their cognitive mechanisms of moral disengagement are likely to be activated. Early study supported the positive relationship between UPB and psychological entitlement (Lee et al., 2017). To sum, drawing on SCT, we have the following hypotheses:

H1 (a) Moral disengagement, (b) organizational identification, (c) identification with supervisor, (d) leader UPB, (e) ethical judgments, (f) perceived forgiveness climate, (g) Machiavellianism, and (h) psychological entitlement positively relate to UPB.

H2 (a) Moral identity, (b) moral courage, (c) taking responsibility, and (d) social desirability negatively relate to UPB.

Social Exchange Theory

Social exchange theory (SET) provides another theoretical lens to understand the antecedents of UPB. The core essential of SET is reciprocity, suggesting that both employees and their organization should benefit from the positive exchange relationship (Blau, 2017). Specifically, when the organization provides support and resources for its employees, the employees may engage in pro-organizational behaviors to reciprocate the initiating action of the organization. Based on SET, we argue that employees may conduct UPB to reciprocate organizational support and care. Early studies found that perceived organizational support (Wang et al., 2021), LMX (Lee et al., 2017), transformational leadership (Effelsberg et al., 2014), benevolent leadership (Shaw & Liao, 2020), organizational commitment (Grabowski et al., 2019), job satisfaction (Dou et al., 2019), and positive reciprocity belief (Umphress et al., 2010) are positively related to UPB.

Perceived Organizational Support

According to SET, when employees received support from their organization, they are likely to engage in pro-organizational behaviors to pay back for the grace (e.g., high levels of OCB and task performance; Cropanzano & Mitchell, 2005). A meta-analysis showed that perceived support from the organization is positively related to employee task performance and contextual performance (Riggle et al., 2009). Similar to task performance and contextual performance, UPB is also a type of behavior that is beneficial to organizational interests although it is against with ethics. Thus, it is reasonable to argue that employees who receive high levels of support from the organization are likely to reciprocate the organization in a manner of UPB.

LMX

Leader–member exchange (Liden & Maslyn, 1998), which reflects the social exchange quality between a leader and her or his followers, may also be an antecedent of UPB based on SET. Specifically, employees with a high level of LMX tend to maintain a good relationship with their leaders and engage in pro-organizational behavior. For them, satisfying the leader's requirements which is useful to maintain a high-quality relationship with the leader is the priority. Considering that leader's requirements are usually consistent with the organizational benefits, employees who pursue a higher level of LMX are more likely to do pro-organizational behaviors even such behaviors are unethical.

Transformational Leadership and Benevolent Leadership

Some positive leadership styles, such as transformational and benevolent leadership, which can activate the followers' reciprocal belief, may also induce their UPB. For example, transformational leadership that highlights individualized consideration for followers (Bass, 1999) is likely to enable the followers to conduct exchange behaviors to render back for the leader's careful consideration; such behaviors are usually beneficial to the organization, but may also go against with ethics. Similar to the above logic, benevolent leadership which highlights a holistic care to followers' well-being (Farh & Cheng, 2000) is also likely to activate the followers' reciprocal belief, which then increases the possibility of the occurrence of UPB.

Organizational Commitment

When employees have a high level of organizational commitment, that is, a force that binds an individual to a target (Meyer & Herscovitch, 2001), they usually develop a strong reciprocal relationship with their organization, because they have a continuance tendency to work in the organization for a long time (Cropanzano & Mitchell, 2005; Cropanzano et al., 2017). Also, employees with a high level of affective commitment usually have a strong motivation to conduct pro-organizational behaviors to benefit the organization. They are more likely to do UPB, even they know such behavior is unethical.

Job Satisfaction

We argue that job satisfaction which reflects an individual's overall job attitude is also a potential antecedent of UPB. Employees with a relatively higher level of job satisfaction usually have received several preferential treatments from the organization (e.g., promotion, high salary, and abundant resource and support; Dou et al., 2019); thus, they may form a stronger reciprocal belief and pro-organizational motivation to conduct pro-organizational behaviors according to SET (Cropanzano & Mitchell, 2005).

Positive Reciprocity Belief

Positive reciprocity belief, which reflects the degree to which an individual places importance on positive reciprocity within exchange relationships (Eisenberger et al., 2004), may serve as a booster of UPB. In a social exchange relationship, people who hold a high level of positive reciprocity belief are more likely to pay back for the benefactor's support and help, to maintain long-term gains for the exchange partner (Umphress et al., 2010). Following this logic, employees with a higher level of positive reciprocity belief are more sensitive to organizational support and grace, and are more likely to conduct pro-organizational behaviors even if such behaviors are unethical. Together, we propose the following hypotheses:

H3 (a) Perceived organizational support, (b) LMX, (c) transformational leadership, (d) benevolent leadership, (e) organizational commitment, (f) affective commitment, (g) job satisfaction, and (h) positive reciprocity belief positively related to UPB.

Social Cognition Theory and Social Exchange Theory

As mentioned in the section of the Introduction, we identify three leadership styles (i.e., ethical leadership, abusive supervision, and authoritarian leadership) that may have double-edged sword effects on UPB.

Ethical Leadership

Ethical leadership, defined as "the demonstration of normatively appropriate conduct through personal actions and interpersonal relationships, and the promotion of such conduct to followers through two-way communication, reinforcement, and decision-making" (Brown et al., 2005, p 120), has also been regarded as a critical antecedent of UPB. However, the relationship between ethical leadership and UPB is still inconsistent. A plausible explanation is that two opposite mechanisms may coexist when investigating the relationship between ethical leadership and UPB. Specifically, ethical leadership may be positively related to UPB via LMX based on SET but may also inhibit UPB through the mechanism of moral disengagement based on SCT.

Drawing on SCT, we have developed the hypotheses between moral disengagement and UPB (see H1), meanwhile, drawing on SET, we have developed the hypotheses between LMX and UPB (see H3). Thus, we would focus on developing the hypotheses of ethical leadership on moral disengagement and LMX. On the one hand, ethical leaders may have an impact on their followers' ethical cognition through their leadership behaviors (e.g., two-way communication and reinforcement). Ethical leaders not only adhere to a high level of ethical standard by themselves but also encourage their followers to comply with such standard. Influenced by these activities, the followers find it very hard to (1) justify their unethical behavior because their role model (i.e., the leader) never conduct such behavior; and (2) minimize or ignore the consequences of unethical behavior because the leader may punish their unethical activities (Brown et al., 2005). It seems that the moral disengagement of employees is unlikely to be activated under the influence of ethical leadership. A meta-analysis also provides evidence to support the negative relationship between ethical leadership and the followers' moral disengagement (Ogunfowora et al., 2021).

On the other hand, equipped with a series of virtues (e.g., honesty, fairness, and altruism), ethical leaders are less likely to abuse their followers even though the followers do not follow their ethical principles; instead, ethical leaders usually treat their followers fairly and considerately, and tend to build a positive social exchange relationship with them (Brown et al., 2005). A meta-analysis also showed that ethical leadership is positively related to LMX (Hoch et al., 2018). To reciprocate the leader's care and support, the followers usually engage in pro-organizational behaviors, even if some of them are unethical, and they will likely not be abused. To sum up, the above two mechanisms give a contradictory explanation to the relationship between ethical leadership and UPB. Specifically, ethical leadership inhibits the followers' moral disengagement process, which then reduces their UPB, while increasing LMX, which may promote UPB. Therefore, we hypothesize the following:

H4 Ethical leadership has a negative relationship with UPB through the mediating effect of moral disengagement.

H5 Ethical leadership has a positive relationship with UPB through the mediating effect of LMX.

Abusive Supervision

Abusive supervision—sustained displays of hostile, nonphysical verbal, and nonverbal behaviors (Tepper, 2000)—may influence the process of moral disengagement. First, abusive supervision gives subordinates opportunities to justify their unethical behavior (Valle et al., 2019), as they have to engage in unethical behavior to avoid abusive behavior by abusive leaders. Second, abusive leaders may pass on a loose moral standard to their followers (Greenbaum et al., 2016). As such, engaging in unethical behavior may not violate the loose ethical standard set up by the abusive leader. An early study has found that abusive supervision is positively related to moral disengagement (Ogunfowora et al., 2021).

Abusive leaders may build a negative exchange relationship with their followers. First, abusive leaders tend to treat their followers in a disingenuous and unethical manner (e.g., invading privacy and lying; Tepper, 2000), which triggers the disgust of the followers. Second, abusive leaders may break their promises for followers (Tepper, 2000), which harms their followers' trust in them and then breaks their relationships with the followers. Third, employees always expect their leaders to demonstrate empathy and provide friendly support to them (Huang et al., 2008). As abusive leaders are less likely to exhibit such behaviors (Xu et al., 2012), employees may not build a positive exchange relationship with their leaders. Consistent with the above arguments, a meta-analysis supported the negative relationship between abusive supervision and LMX (Mackey et al., 2015). Together, abusive supervision may positively relate to UPB via moral disengagement and negatively influence UPB via LMX simultaneously. Thus, we propose the following hypotheses:

H6 Abusive supervision has a positive relationship with UPB through the mediating effect of moral disengagement.

H7 Abusive supervision has a negative relationship with UPB through the mediating effect of LMX.

Authoritarian Leadership

Authoritarian leadership, emphasizing rigorous control over subordinates and the subordinates' obedience for the leader (Farh & Cheng, 2000), may influence UPB positively via moral disengagement and negatively via LMX. To start, under the influence of authoritarian leadership, employees cannot express personal views, they could only achieve the performance goals set by their leaders (Liu et al., 2021). Employees may justify their unethical behavior on the grounds that they have to achieve performance goals. Besides, the leader is usually regarded as the moral model for their followers. When leaders conduct unethical behavior (i.e., authoritarian behavior) and do not need to take responsibility, their followers are likely to believe that they could engage in unethical behavior without taking responsibility fully and thereby trigger moral disengagement. Previous research also showed that authoritarian leadership is positively related to moral disengagement (Shaw et al., 2020).

Then, authoritarian leadership may be negatively related to LMX. First, authoritarian leaders tend to control information, ignore followers' contributions, attribute losses to subordinates, and punish them (Guo et al., 2018), all of which may destroy the relationships between their followers and them. Second, authoritarian leaders are likely to trigger their followers' negative affect (e.g., fear and anger) toward them, which may damage the quality of their affective relationship (Gu et al., 2018). Third, considering that low-quality relationships are usually defined as hierarchical and role-defined (Bedi, 2019) and authoritarian leaders always exert hierarchical and controlling behaviors to monitor their followers (Farh & Cheng, 2000), they may build low-quality relationships with their followers. Consistent with the above arguments, a meta-analysis found a negative linkage between authoritarian leadership and LMX (Hiller et al., 2019). Therefore, we hypothesize the following:

H8 Authoritarian leadership has a positive relationship with UPB through the mediating effect of moral disengagement.

H9 Authoritarian leadership has a negative relationship with UPB through the mediating effect of LMX.

Demographic Variables

In addition to social influences, we also identified four demographic variables (i.e., age, tenure, gender, and education level) as the antecedents of UPB.

Age and Tenure

Age and tenure may hinder UPB. In the organizational context, conducting unethical behavior is risky for employees, as they are likely to be blamed or even punished by supervisors, peers, consumers, or the public. Considering that younger individuals are more likely to take risks than older ones (Defoe et al., 2015), age should be negatively associated with UPB. Similar to the logic, employees with relatively long work tenure are less likely to take risks. Moreover, experienced employees tend to be more cautious about conducting unethical behaviors, as they are sensitive to the loss of their current status and acquired recognition (Byrne & Anaraky, 2020). Based on the above arguments, tenure should be negatively related to UPB. An early study confirmed that age and tenure are negatively related to UPB (Umphress et al., 2010).

Education Level

Individuals with higher education levels are less likely to engage in UPB. The higher the level of education means the higher the level of human capital investment (Becker, 1962). With a higher level of human capital investment, the cost of UPB would increase, decreasing the intention to engage in UPB. Early studies found that education level was negatively related to UPB (Lee et al., 2020; Tang & Li, 2021).

Gender

Gender differences may exist in reporting UPB. Men and women have different social expectations. Men are expected to achieve more competitive success (i.e., money and advancement) than women (Betz et al., 1989). Thus, male employees may find a reasonable excuse to do UPB, as they should achieve high expectations and competitive success. Moreover, a previous study found that women are more likely to perceive hypothetical business practices as unethical (Franke et al., 1997). As such, women may hold a higher level of moral standard in business practice than men, and thereby, reporting less UPB. Previous studies also found that men reported more UPB than women (Lian et al., 2020; Tang & Li, 2021). To sum, we hypothesize:

H10 Gender (0 = female, 1 = male) has a positive relationship with UPB.

H11 (a) age, (b) tenure, and (c) education level negatively relate to UPB.

Method

Literature Search

To be as inclusive as possible, we use multiple strategies to identify studies that include relationships between UPB and its antecedents. Specifically, a search was conducted in September 2021 in the following databases: PsycINFO, Web of Science, and Google Scholar. An updated search was conducted in March 2022 in the Review Process. Keywords were determined according to Umphress et al. (2010) since they developed the first measurement scale of UPB. We use keywords such as unethical pro-organizational behavior and UPB to locate all potential studies about UPB.

Inclusion Criteria and Exclusion Criteria

Several decision rules were used to decide which studies should be included or excluded. Studies are included in this meta-analysis based on the following criteria: (a) being written in English; (b)sampling from the workplace; and (c) reporting effect sizes (correlations). Studies are excluded because: (a) not measuring UPB, for example, only measuring unethical behavior or counter-productive work behavior; and (b) not including correlations that we are interested in.

Coding

To ensure the quality of coding, two authors independently coded the following information: bibliographic references (authors and publication year), sample size, research design/sampling strategy, effect sizes (correlations), and the reliabilities of all scales. The average intercoder percentage of agreement across the study variables was 97%. Differences were discussed until a consensus was reached. Finally, we got 296 independent correlations from 67 papers including 20,957 samples. A PRISMA flowchart was provided (see Fig. 2).

Analytical Approach

To get the true-score correlations between UPB and its antecedents, we used the Hunter–Schmidt method random-effect model meta-analysis. Measurement error was corrected using reliability (Hunter & Schmidt, 2004). When reliabilities were not reported, we used sample-weighted reliabilities (Ogunfowora et al., 2021; Young et al., 2021). *Psychmeta* package (Dahlke & Wiernik, 2019) in R was used to conduct the results of the meta-analysis. We also tried to detect publication bias drawing on Egger's regression method using *metafor* package (Viechtbauer, 2010) in R. The results are shown in Table 1.

Path Analysis

MASEM technology is used to study the double-edged sword effects of ethical leadership, abusive supervision, and authoritarian leadership on UPB. First, we built three meta-analytic correlations matrices for path analysis. We tried to find all correlations that these matrices need from the current and the previous meta-analysis. Unfortunately, we could not locate the correlations between LMX and moral disengagement; thereby, we made an original meta-analysis to calculate their true-score correlation. Specifically, we searched the studies that included their correlations drawing on keywords: "moral disengagement," "LMX," and "leader-member exchange" from the databases we mentioned above. Then, we coded the necessary information and used a random-effect model to calculate their true-score correlation. Drawing on the matrices we built, we used Mplus to accomplish path analysis. Following the guidance by Viswesvaran & Ones (1995), the harmonic mean sample size was used in path analysis. Path coefficient, 95% CI, and p value were created by MPLUS. The results are shown in Figs. 3, 4, and 5 and Tables 2, 3, 4, 5, 6, and 7.

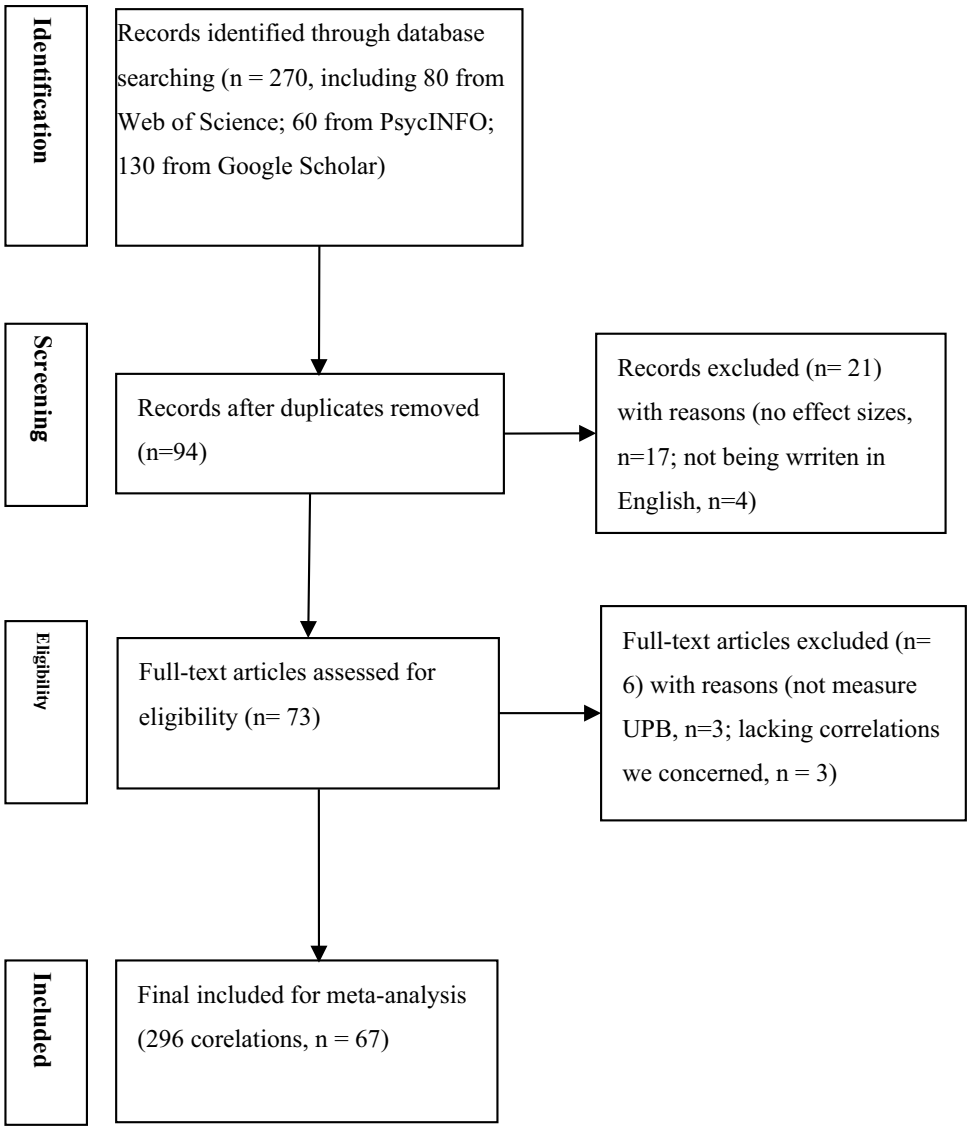


Fig. 2 PRISMA flowchart

Results

First, moral disengagement (ρ = 0.51), organizational identification (ρ = 0.26), identification with supervisors (ρ = 0.23), leader UPB (ρ = 0.45), ethical judgments (ρ = 0.47), moral identity (ρ = − 0.21), taking responsibility (ρ = − 0.32), transformational leadership (ρ = 0.20), and job satisfaction (ρ = 0.21), psychological entitlement (ρ = 0.51), and gender (ρ = 0.05) significantly related to UPB (95%CI did not include zero, see Table 1). In particular, for gender, ρ = 0.05 > 0 means males reported higher UPB than females. Perceived forgiveness climate, moral courage, perceived organizational support, LMX, benevolent leadership, education, organizational commitment, affective commitment, and positive reciprocity belief, Machiavellianism, social desirability, age, and tenure are insignificantly related to UPB (95%CI include zero, see Table 1). Thus, H1(a), H1(b), H1(c), H1(d), H1(e), H1(h), H2(a), H2(c), H3(c), H3(g), and H10 were supported; H1(f), H1(g), H2(b), H2(d), H3(a), H3(b), H3(d), H3(e), H3(f), H3(h), H11(a), H11(b), and H11(c) were rejected. Besides, we noticed the majority of *p* values of Egger's regression are not significant, showing that publication bias is not serious in the current study. For instance, in relation to affective commitment, the *p* value is equal to 0.9 which is larger than 0.05.

Second, for the double-edged effects of ethical leadership on UPB, from Table 3, we noticed that moral disengagement negatively mediated the link between ethical leadership and UPB [(estimate = − 0.07, 95%CI = (− 0.09, − 0.05)], while LMX positively mediated the link between ethical leadership and UPB [estimate = 0.15, 95%CI = (0.12, 0.17)]. Thus, H4 and H5 were supported.

Third, for abusive supervision, from Table 5, we found that moral disengagement positively mediated the link

Table 1 Results of antecedents of UPB

variable	*k*	*N*	r	ρ	SDρ	95% CI	80% CR	Egger's regression	
								t	*p*
Abusive supervision	3	1043	0.03	0.03	0.35	[−0.86, 0.91]	[−0.64, 0.69]	0.21	0.87
Affective commitment	5	1447	0.11	0.12	0.23	[−0.18, 0.42]	[−0.24, 0.48]	0.13	0.9
Age	51	14,003	−0.02	−0.02	0.07	[−0.05, 0.01]	[−0.11, 0.07]	−1.16	0.25
Authoritarian leadership	2	377	0.18	0.2	0.13	[−1.20, 1.61]	[−0.21, 0.62]	–	–
Benevolent leadership	2	432	0.11	0.12	0.02	[−0.57, 0.82]	[0.07, 0.18]	–	–
Education	24	6 983	−0.02	−0.02	0.08	[−0.07, 0.02]	[−0.12, 0.08]	−1.96	0.06
Ethical judgments	3	456	0.43	0.47	0	[0.41, 0.53]	[0.47, 0.47]	7.45	0.09
Ethical leadership	6	1493	−0.00	−0.01	0.12	[−0.15, 0.14]	[−0.18, 0.17]	0.56	0.61
Gender	50	13 746	0.05	0.05	0.08	[0.02, 0.08]	[−0.05, 0.15]	0.28	0.78
Identification with supervisors	8	1836	0.2	0.23	0.15	[0.10, 0.37]	[0.02, 0.44]	1.21	0.27
Job satisfaction	4	1409	0.18	0.21	0	[0.15, 0.27]	[0.21, 0.21]	0.39	0.74
Leader UPB	4	803	0.38	0.45	0.17	[0.15, 0.74]	[0.17, 0.73]	0.4	0.73
LMX	8	2191	0.14	0.17	0.22	[−0.03, 0.36]	[−0.15, 0.47]	−0.51	0.63
Machiavellianism	4	696	0.24	0.26	0.36	[−0.33, 0.85]	[−0.33, 0.85]	1.31	0.32
Moral courage	2	366	−0.06	−0.09	0.14	[−1.73, 1.54]	[−0.54, 0.35]	–	–
Moral disengagement	9	1977	0.43	0.51	0.22	[0.33, 0.69]	[0.20, 0.82]	2.14	0.07
Moral identity	14	3542	−0.18	−0.21	0.22	[−0.34, −0.08]	[−0.50, 0.08]	−0.41	0.69
Organizational commitment	2	729	0.24	0.27	0.15	[−1.18, 1.72]	[−0.19, 0.74]	–	–
Organizational identification	25	6058	0.22	0.26	0.19	[0.17, 0.34]	[0.01, 0.51]	−2.3	0.03
Perceived organizational support	2	354	0.1	0.11	0	[−0.35, 0.58]	[0.11, 0.11]	–	–
Perceived forgiveness climate	2	634	−0.09	−0.10	0.44	[−4.12, 3.91]	[−1.46, 1.26]	–	–
Positive reciprocity beliefs	4	1173	0.17	0.19	0.19	[−0.13, 0.51]	[−0.13, 0.50]	0.62	0.65
Psychological entitlement	5	1 247	0.46	0.51	0.26	[0.18, 0.84]	[0.12, 0.91]	−0.67	0.55
Social desirability	14	3144	0.06	0.07	0.23	[−0.07, 0.21]	[−0.24, 0.37]	1.34	0.21
Taking responsibility	2	366	−0.23	−0.32	0	[−0.52, −0.12]	[−0.32, −0.32]	–	–
Tenure	37	10,232	−0.02	−0.02	0.09	[−0.06, 0.02]	[−0.14, 0.10]	−0.87	0.39
Transformational leadership	4	935	0.18	0.2	0.07	[0.04, 0.37]	[0.08, 0.32]	0.35	0.76

k number of studies contributing to meta-analysis, N total sample size, r mean observed correlation, ρ mean true-score correlation, SDρ standard deviation for population estimate

CI confidence interval, *CR* credibility interval

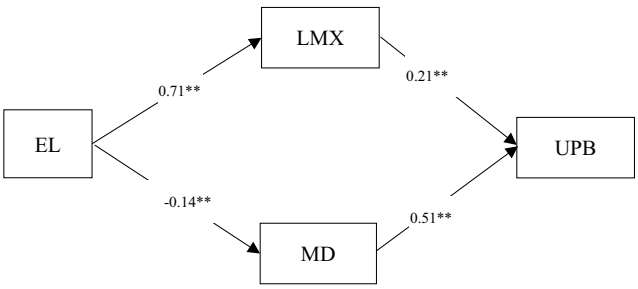


Fig. 3 Estimated mediation model for ethical leadership

Note: * $p < 0.05$; ** $p < 0.01$. EL represents ethical leadership; LMX represents leader-member exchange; MD represents moral disengagement; UPB represents unethical pro-organizational behavior.

between abusive supervision and UPB [estimate = 0.29, 95%CI = (0.26, 0.33)], while LMX negatively mediated the link between abusive supervision and UPB

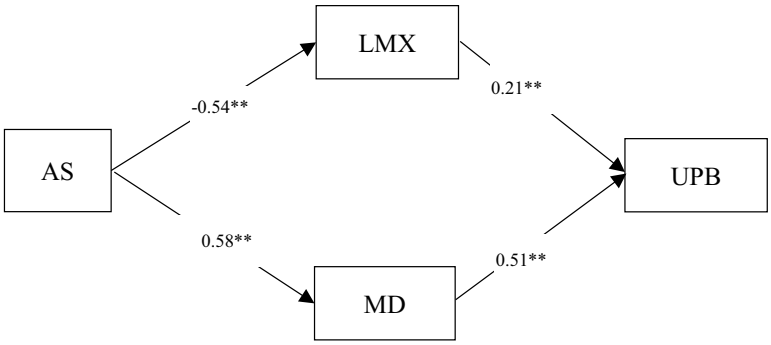


Fig. 4 Estimated mediation model for abusive supervision

Note: * $p < 0.05$; * * $p < 0.01$. AS represents abusive supervision.

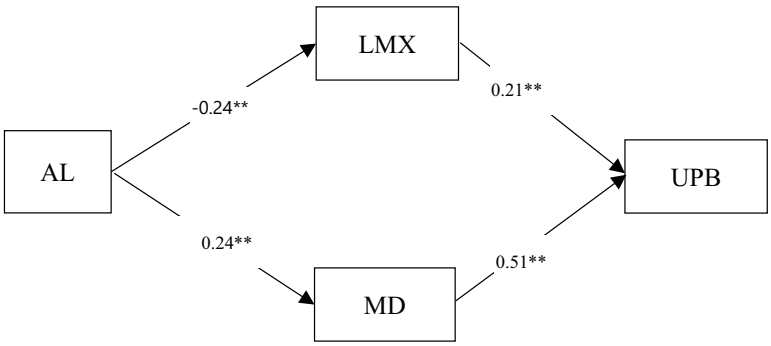


Fig. 5 Estimated mediation model for authoritarian leadership

Note: * $p < 0.05$; * * $p < 0.01$. AL represents authoritarian leadership.

Table 2 Meta-analytic correlation matrix for ethical leadership

		1	2	3	4
1	EL	1			
	K(N)				
2	LMX	0.71[a]	1		
	K(N)	18(4052)			
3	MD	−0.14	0.02	1	
	K(N)	17(6038)[b]	4(900)		
4	UPB	−0.01	0.22	0.51	1
	K(N)	6(1493)	8(2191)	9(1977)	

EL Ethical leadership, *MD* Moral disengagement, unlabeled correlations came from the current meta-analysis

[a]Hoch et al. (2018)

[b]Ogunfowora et al. (2021)

Table 3 Coefficients for mediation model for ethical leadership

Path	Estimate	SE	95%CI	*p* value
EL → MD	−0.14	0.02	[−0.18, −0.10]	< 0.001
EL → LMX	0.71	0.02	[0.68, 0.74]	< 0.001
MD → UPB	0.51	0.02	[0.47, 0.54]	< 0.001
LMX → UPB	0.21	0.02	[0.17, 0.25]	< 0.001
indirect effect				
EL → MD → UPB	−0.07	0.01	[−0.09, −0.05]	< 0.001
EL → LMX → UPB	0.15	0.01	[0.12, 0.17]	< 0.001

N = 1902 (Harmonic mean), Chi-Square = 102.19, df = 2, $p < 0.001$, CFI 0.95, SRMR = 0.06

[estimate = − 0.11, 95%CI = (− 0.14, − 0.09)]. Therefore, H6 and H7 were supported.

Fourth, for the authoritarian leadership, from Table 7, we revealed that moral disengagement positively mediated the link between authoritarian leadership and UPB [(estimate = 0.12, 95%CI = (0.08, 0.16)], while LMX negatively mediated the link between authoritarian leadership and UPB [estimate = − 0.05, 95%CI = (− 0.07, − 0.03)]. As such, H8 and H9 were supported.

Discussion

To our knowledge, our meta-analysis is the first quantitative review of the antecedents of UPB. We collected 296 correlations from 67 independent studies to systematically research the antecedents of UPB. Our meta-analysis had two primary goals. The first one is to accurately estimate

Table 4 Meta-analytic correlation matrix for abusive supervision

		1	2	3	4
1	AS	1			
	K(N)				
2	LMX	−0.54[a]	1		
	K(N)	11(2786)			
3	MD	0.58	0.02	1	
	K(N)	3(2270)[b]	4(900)		
4	UPB	0.03	0.22	0.51	1
	K(N)	3(1043)	8(2191)	9(1977)	

AS Abusive supervision

[a]Mackey et al. (2015)

[b]Ogunfowora et al. (2021)

Table 5 Coefficients for mediation model for abusive supervision

Path	Estimate	SE	95%CI	*p* value
AS → MD	0.58	0.02	[0.54, 0.62]	< 0.001
AS → LMX	−0.54	0.02	[−0.58, −0.50]	< 0.001
MD → UPB	0.51	0.02	[0.46, 0.55]	< 0.001
LMX → UPB	0.21	0.02	[0.17, 0.25]	< 0.001
indirect effect				
AS → MD → UPB	0.29	0.01	[0.26, 0.33]	< 0.001
AS → LMX → UPB	−0.11	0.01	[−0.14, −0.09]	< 0.001

N = 1566 (Harmonic mean), Chi-Square = 569.67, df 2, $p < 0.001$, CFI = 0.76, SRMR = 0.14

Table 6 Meta-analytic correlation matrix for authoritarian leadership

		1	2	3	4
1	AL	1			
	K(N)				
2	LMX	-0.24[a]	1		
	K(N)	8(3274)			
3	MD	0.24	0.02	1	
	K(N)	2(373)	4(900)		
4	UPB	0.2	0.22	0.51	1
	K(N)	2(377)	8(2191)	9(1977)	

AL Authoritarian leadership

[a]Hiller et al. (2019)

the relationship between UPB and its antecedents, making a methodological contribution. The second one is to test the double-edged effects of ethical leadership, abusive supervision, and authoritarian leadership on UPB, making a theoretical contribution. In the following, we will discuss these findings according to our research framework and then propose some future research directions.

Table 7 Coefficients for mediation model for authoritarian leadership

Path	Estimate	SE	95%CI	*p* value
AL → MD	0.24	0.04	[0.17, 0.30]	< 0.001
AL → LMX	−0.24	0.04	[−0.31, −0.17]	< 0.001
MD → UPB	0.51	0.03	[0.45, 0.57]	< 0.001
LMX → UPB	0.21	0.03	[0.15, 0.26]	< 0.001
Indirect effect				
AL → MD → UPB	0.12	0.02	[0.08, 0.16]	< 0.001
AL → LMX → UPB	−0.05	0.01	[−0.07, −0.03]	< 0.001

N = 778 (Harmonic mean), Chi-Square = 26.66, df 2, $p < 0.001$, CFI = 0.94, SRMR = 0.05

SCT

Guiding by the standard that small effect sizes are correlations of 0.10, moderate are 0.30, and large are 0.50 (Cohen, 2013), we discuss the magnitude of effect sizes.

First, we found moral disengagement (ρ = 0.51), organizational identification (ρ = 0.26), identification with supervisors (ρ = 0.23), leader's UPB (ρ = 0.45), ethical judgments (ρ = 0.47), moral identity (ρ = −0.21), psychological entitlement (ρ = 0.51), and taking responsibility (ρ = −0.32) were significantly related to UPB. Because all of these antecedents were closely related to employees' ethical cognition and were identified based on SCT, our results indicated that SCT should be regarded as an important theoretical perspective to explain antecedents of UPB. In addition, it is worthy to mention that moral disengagement (ρ = 0.51) and psychological entitlement (ρ = 0.51) are largely related to UPB. As such, they should serve as strong and proximal predictors of UPB.

Second, out of our expectations, perceived forgiveness climate, Machiavellianism, moral courage, and social desirability were not significantly related to UPB. One plausible explanation is that the number of studies included in the meta-analysis is few (e.g., $k = 2$, for perceived forgiveness climate; $k = 2$, for moral courage). When k is low, the error variance across studies could be large (Hunter & Schmidt, 2004), causing the confidence interval to be broad including zero. As such, more primacy studies are called to accurately estimate their links.

Third, we did not find a significant relationship between Machiavellianism and UPB. Machiavellians' moral principles are contingent. Only in the condition where they can benefit themselves from UPB, they will do it (Castille et al., 2016). That is, when doing UPB does not benefit Machiavellians, they may not engage in UPB. On the other hand, they usually lack empathy (Paal & Bereczkei, 2007), implying that they are less likely to engage in pro-organizational behavior.

Finally, social desirability is also nonsignificantly related to UPB. A possible explanation is that people with high

social desirability tend to avoid acknowledging the unethical part of UPB, but may report more pro-organizational behavior at the same time. Thus, as our meta-analysis revealed, the correlation between social desirability and UPB is close to zero.

SET

First, we found that transformational leadership is significantly positively related to UPB, while benevolent leadership and LMX are nonsignificantly related to UPB. A small number of studies (for benevolent leadership, $k = 2$; for LMX, $k = 7$, see Table 1) were included in the meta-analysis, therefore, it is reasonable to find nonsignificant results regarding their relationships because we cannot effectively correct sampling error drawing on a small number of studies. Previous meta-analyses showed that transformational leadership (Hoch et al., 2018), benevolent leadership (Hiller et al., 2019), and LMX (Rockstuhl et al., 2012) are positively related to OCB. Thus, we theoretically believe that these positive leadership styles may facilitate followers' pro-organizational behavior. When more empirical evidences are accumulated, we call for an updated meta-analysis to evaluate the accurate relationships between benevolent leadership (LMX) and UPB.

Second, we found that job satisfaction was positively related to UPB, while organizational commitment and affective commitment are nonsignificantly related to UPB. Similarly, the small number of studies included in our meta-analysis ($k = 2$, for organizational commitment; $k = 5$, for affective commitment) may be a possible reason for these nonsignificant results. Considering that previous meta-analyses revealed positive linkages between organizational commitment, affective commitment (Meyer et al., 2002), and job satisfaction (Fassina et al., 2007) and OCB, we speculated that these variables may be positively related to the pro-organizational part of UPB. The unsolved question is that whether employees with high levels of organizational or affective commitment will conduct pro-organizational behavior even if these behaviors go against with ethics. We encourage future research to address this issue.

Finally, we found that perceived organizational support and positive reciprocity beliefs are nonsignificantly related to UPB. Likewise, we attributed the nonsignificant results to the limited number of empirical studies regarding the relationships between perceived organizational support (positive reciprocity beliefs) and UPB. Due to the small number of studies and broad confidence interval, the correlations with them should be interpreted cautiously.

To sum, some of the antecedents guided by SET were nonsignificantly related to UPB. These findings are inconsistent with our theoretical expectations. The main reason is that we have quite limited empirical evidences tapping SET. Considering the validity of SET on explaining pro-organizational behavior, we encourage more empirical studies to focus on how to adopt SET to explore the antecedents of UPB.

SCT and SET

We found ethical leadership, abusive supervision, and authoritarian leadership have double-edged effects on UPB.

First, we found that moral disengagement negatively mediated the link between ethical leadership and UPB, and that LMX positively mediated the linkage between ethical leadership and UPB. Our research unpacked the black box between ethical leadership and UPB, and further reconciled the previous inconsistent findings between ethical leadership and UPB. Because of the existence of two opposite mechanisms, the correlation between ethical leadership and UPB is close to zero (see Table 1, ρ between ethical leadership and UPB is −0.01).

Second, moral disengagement positively mediated the link between abusive supervision (authoritarian leadership) and UPB, and LMX negatively mediated the linkage between abusive supervision (authoritarian leadership) and UPB. As such, not only ethical leadership but also abusive supervision and authoritarian leadership have double-edged effects on UPB. Interestingly, do other leadership styles have such double-edged effects on UPB? Since no survey studies researched this topic, we could not address this question drawing on MASEM. It seems that leadership styles may influence UPB through both LMX and moral disengagement. For example, transformational leadership may influence UPB via LMX and moral disengagement at the same time. As Table 1 shows, the ρ between transformational leadership and UPB is 0.2, implying that the two mechanisms might both be positive. Our findings are valuable, because they inspire scholars to take the unethical and the pro-organizational components into consideration simultaneously when exploring the leadership antecedents of UPB, and also provide empirical evidence to justify SCT and SET as valid theoretical lenses to specifically explain the unethical and the pro-organizational components of UPB. Together, when researching the links between leadership and UPB, we need to consider two theoretical perspectives together.

Third, it seems that abusive supervision has a stronger impact on UPB through both LMX and moral disengagement than authoritarian leadership. For example, the estimate of the indirect effect of abusive supervision on UPB through LMX (estimate = −0.11, see Table 5) is stronger than authoritarian leadership (estimate = −0.05, see Table 7). This result may imply that despite they are both negative leadership, abusive supervision may trigger stronger impacts on LMX and moral disengagement than authoritarian leadership. Although authoritarian leaders

control and give pressure employees (Farh & Cheng, 2000), they do not verbally insult their followers. Compared to authoritarian leaders, abusive leaders may inflict more serious negative influences on employees through a variety of intense harming behaviors (e.g., rude behavior and verbal insult, Tepper, 2000).

Finally, as mentioned above, 95% CI between LMX and UPB included zero, but their correlation is also used in our analysis of the mediating effect of LMX. This is acceptable when we adopted MASEM technology to test the mediating effects because building a complete meta-analysis correlation matrix could be hard and sometimes we have to use correlations that only had a few samples and 95% CI included zero (see previous meta-analyses, Young et al., 2021; Chung et al., 2022).

Demographic Antecedents UPB

In a survey study, to control (or not) some demographic variables need an evidence-based reason. The current study seeks to provide evidence to control some demographic variables.

First, we did not find a significant link between education level and UPB. Education may have raised the cost of being punished when unethical behavior is disclosed. However, higher education levels, along with higher human capital and a higher salary, may make UPB more profitable. As such, the correlation between education and UPB is close to zero ($\rho = -0.02$, see Table 1).

Second, a weakly positive relationship has been found between gender and UPB. In particular, men reported more UPB than women. This finding is in line with our expectations. Drawing on our evidence, it is necessary to control gender in the survey study of UPB.

Third, inconsistent with our expectations, age and tenure were nonsignificantly related to UPB. In our hypotheses development, we argue that old and senior employees may engage in less unethical behavior. One possible explanation is that aged people are more likely to do pro-organizational behavior because they usually have a well social exchange relationship with their organizations. For example, a meta-analysis showed age was positively related to OCB (Ng & Feldman, 2008). These two opposite effects lead to a nonsignificant relationship between age (tenure) and UPB. Another explanation is that there may be a curvilinear relationship between age/tenure and UPB. Indirect evidence showed that curvilinear relationships exist between age and performance (Sturman, 2003), risk-taking behavior (Yeoh & Hooy, 2020), and happiness (Blanchflower, 2021). This evidence may suggest that the relationship between age and behavior could be complex and even curvilinear.

Practical Implications

Our findings are useful for managers to systematically understand the antecedents of UPB and provide suggestions for them to adopt proper interventions to reduce employees' UPB. On the one hand, organizations should try to encourage employees to conduct more pro-organizational behaviors. For example, managers can provide social support to employees, build positive social exchange relations with them, and try to increase their job satisfaction. On the other hand, organizations should also take actions to prevent employees' unethical behavior through inhibiting the moral disengagement process of employees. To achieve this goal, ethical leadership may play an important role. For example, ethical managers should clarify the negative influences of UPB, and tell their followers that although UPB may increase the interests of the organization in a short term, it ultimately harms the interest of the organization in a long team. Ethical leaders should also consider the potential negative consequences of UPB when evaluating their followers' work performance. In addition, ethical leaders can also build an ethical climate to restrict their follower's unethical behaviors. Abusive supervision and authoritarian leadership should be avoided, as they would damage the positive exchange relationship between leaders and employees.

Limitations and Future Directions

Our study has several limitations. First, meta-analysis is based on the correlations between variables; thus, we cannot empirically justify the causality from the antecedent variables to UPB. Future studies can try to confirm the causality between UPB and its antecedents by using experimental research design. Second, we only coded the articles written in English, which limited the number of samples included in our meta-analysis. In the future, researchers can try to code the relevant studies written in other languages. Third, because of the limitation of the number of studies included in the meta-analysis, many nonsignificant links of interest in the SET perspective were found. An updated meta-analysis is called when more empirical evidence is accumulated. Finally, we applied some correlations whose 95%CI included zero into path analysis. Although this is acceptable in MASEM, 95%CI included zero means that the estimation of ρ and path coefficient generated by ρ is not so accurate. When more primacy studies are conducted, we encourage future meta-analyses to provide a more accurate estimation of path coefficients.

As a meta-analysis, our goal is not only to summarize what we have already known, but also to generate several promising future directions. To start, the fundamental concern of UPB is the measurement issue. We noticed that the most widely used measurement tool of UPB is the scale developed by Umphress

et al.(2010). This scale, however, captures the intention rather than the behavior of UPB. The intention cannot fully reflect the actual behavior. In addition, future studies should also consider the different contents of UPB in specific industries and jobs. In specific, the scale items should be carefully considered to fit the specific job contents. The item that "If my organization needed me to, I would withhold issuing a refund to a customer or client accidentally overcharged" is more suitable for salesmen, and "If my organization needed me to, I would give a good recommendation on the behalf of an incompetent employee in the hope that the person will become another organization's problem instead of my own" is more proper for HR managers. We encourage future studies to further consider the measurement issue of UPB, and to develop behavior- and contextual-based scales.

Then, we encourage future studies to consider more personal traits as antecedents of UPB. For example, the regulation focus should be seriously considered. Conducting UPB is always associated with high risk and high return. People who have a promotion focus may conduct more UPB because they have a high level of gain-seeking inclinations; meanwhile, people who have a prevention focus may conduct less UPB because they are very sensitive to loss (Higgins, 2000). Moreover, similar to Machiavellianism, some other "dark triads" (e.g., psychopathy) should be considered as the potential antecedent of UPB. Because people high in psychopathy feel less guilty when their actions harm others (Forsyth et al., 2012), they may be more likely to do UPB. Future studies should examine such a relationship.

Next, we encourage to use more theoretical perspectives to study the antecedents of UPB beyond the SCT/SET framework. First, self-determination theory (Deci et al., 2017) is encouraged to explain the antecedents of UPB. For example, it is interesting to explore employees' motivation of engaging in UPB. That is, their UPB is driven by autonomous motivation or controlled motivation. Second, future time perspectives (Kooij et al., 2018) could be taken into consideration. Specifically, individuals with different time perspectives may have differences in engaging in UPB.

Finally, we should pay more attention to the outcomes of UPB in the future. When searching for the related literature, we found that most studies focused on the antecedents of UPB, and very limited studies tapped the consequences of UPB. Because we theoretically argue that employees' UPB may benefit their organization for a short time but harm the long-term interests of the organization, we need more empirical evidence to support this proposition in the future.

Conclusion

In the past decade, UPB has drawn loads of academic attention. To make it more clear about how to prevent UPB, we conducted a meta-analytic review of the antecedents of UPB. Our results indicated that moral disengagement, organizational identification, identification with supervisors, leader UPB, ethical judgments, transformational leadership, psychological entitlement, gender, and job satisfaction are positively related to UPB. We also found that moral identity and taking responsibility are negatively related to UPB. In addition, ethical leadership has a double-edged sword effect on UPB. On the one hand, it increases UPB via LMX, and on the other hand, it reduces UPB via moral disengagement. Abusive supervision and authoritarian leadership also have double-edged sword effects on UPB, decreasing UPB via LMX, while increasing UPB via moral disengagement. Our study provides solid evidence to understand the antecedents of UPB accurately, contributing to UPB literature. We suggest some promising directions for future research on UPB, hoping UPB studies could move forward.

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Declarations

Conflict of interest All the authors have no potential conflicts of interest to disclose.

Research Involving Human Participants and/or Animals This study is a meta-analytic review, and we did not collect data from any human participants and/or animals.

Informed Consent Because our study is a meta-analytic review and we did not collect data from any human participants and/or animals, we do not have informed consent.

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*Coding information and papers included in the current meta-analysis are shown in the supplemental materials.

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